1. Introduction

1.1 Background

In Ethiopia, many restaurants still use traditional methods to **manage orders** and **serve customers**. Customers usually need to visit the restaurant in person, and place their orders manually. During busy times, this often causes long queues, delays, and inefficient service.

Although internet access and smartphone use are increasing, many restaurants have not adopted modern digital systems for ordering food, reserving tables, or managing deliveries. In addition, unstable internet connections in some areas make online services difficult to use consistently.

Restaurants also face challenges when handling many customers at once. Staff must manage orders, tables, and communication with the kitchen at the same time, which can lead to mistakes and slow service.

1.2 Problem Statement

Food delivery and restaurant service systems in Ethiopia still experience several challenges. Customers often face delays, long waiting times, and limited information about restaurant availability. Many platforms do not function well in areas with weak internet connectivity, which prevents users from accessing menus or placing orders smoothly.

Restaurants also struggle to manage high-demand periods, organize queues, and prioritize important orders. Payment processes may lack transparency, and customers cannot always track their orders or know when their food will be ready.

Because of these issues, digital food services are not as efficient or widely trusted as they could be. There is a need for a system that improves order management, supports offline access, provides clear communication between users and restaurants, and integrates local payment methods suitable for the Ethiopian environment.

1.3 General Objective

The general objective of this project is to design and develop a **modern**, **responsive**, and **weak Connection capable** food delivery system that improves customer experience, optimizes restaurant order management, and provides secure and reliable payment options.

1.4 Specific Objectives

- Develop a Progressive Web Application (PWA) that works across desktop and mobile devices.
- Enable customers to browse restaurants, view menus, and place orders easily.
- Allow restaurants to manage orders, reservations, and customer queues digitally.
- Provide real-time notifications for customers, restaurants, and delivery riders.
- Support weak connection functionality for users with unstable internet connections.
- Integrate multiple payment methods including cash and local digital payment platforms.
- Support local languages for native users,
- Implement order tracking and estimated delivery time features.
- Provide a rating and review system for customer feedback.
- Ensure secure handling of user data and transactions.

1.5 Scope of the System

- This system will focus on providing a digital platform for food ordering, restaurant management, and delivery coordination within Ethiopian City and Sub cities.
- Users will be able to access the system through a web application designed to function like a mobile app using Progressive Web Application technology.
- The system will also support integration with local payment services.

Out scope

The initial version of the system will not include native mobile applications or complex machine learning systems.

1.6 Significance of the System

General Significance

This system helps to improve how food ordering and delivery work in Ethiopia by making it easier for customers, restaurants, and delivery riders to connect through a digital platform.

Specific Significance

- **Saves Time for Customers**: Customers can view menus, order food, and reserve tables without traveling to the restaurant first.
- **Reduces Waiting in Restaurants**: The system allows customers to know when to come or when their food will be ready, which helps reduce long queues.
- **Helps Restaurants Work More Efficiently**: Restaurants can manage orders, tables, and customer requests in one system instead of using paper or memory.
- **Works Even with Weak Internet**: Some features will still work when the internet connection is not strong.
- **Supports Local Payment Methods**: Customers can pay using local digital payment systems.
- **Improves Delivery Service**: Delivery riders receive clear order information, which helps deliver food faster and get income.
- **Helps Restaurants Improve Their Service**: Customer ratings and reviews help restaurants understand what customers like and what needs improvement.

1.7 Targeted Beneficiaries

Primary Beneficiaries

Restaurants

Restaurants are the main beneficiaries of this system. The platform helps them manage their operations more efficiently and serve more customers.

Benefits for Restaurants

- **More Customers**: Restaurants can reach customers who prefer ordering online instead of visiting physically.
- **Better Order Management**: Orders are organized in a digital system, reducing mistakes and confusion during busy hours.
- **Reduced Waiting Time**: Table reservations and scheduled orders help control crowding and improve service flow.
- **Higher Revenue Opportunities**: With delivery and online ordering, restaurants can sell more food beyond walk-in customers.
- **Improved Customer Satisfaction**: Faster service and clear communication increase customer trust and loyalty.
- **Customer Insights**: Reviews, ratings, and order data help restaurants understand what customers like most.
- **Efficient Queue Management**: The system helps restaurants manage high demand and prioritize important orders when necessary.

Secondary Beneficiaries

Customers: Customers benefit from **faster ordering**, **less waiting time**, **the ability to reserve tables**, and the convenience of **ordering from anywhere**.

Delivery Riders: Delivery riders **receive clear delivery requests**, **order details**, and **directions**, helping them work more efficiently and get income.

Local Food Industry: The system encourages digital adoption in the restaurant industry and helps improve overall service quality.

1.8 Feasibility Study

1. Technical Feasibility

The system is technically feasible because it will be built using modern and widely supported web technologies. These technologies are **reliable, scalable,** and **suitable** for building a **Progressive Web Application** (PWA).

The proposed development stack includes:

- **Frontend:** TypeScript
- **Backend:** Node.js with Express
- **Database:**

- **ORM:** Prisma
- **Offline Support:** Service Workers and Workbox
- **Deployment Platforms:** Cloud services that offer free or low-cost hosting

2. Economic Feasibility

- The system can be developed and deployed with **minimal initial cost**.
- Many of the tools and technologies used in the project are **open-source and free to use**.
- During the initial stage, the system can be **hosted using free or low-cost cloud platforms for testing and early deployment**.
- Restaurants will not need special hardware; they can use existing smartphones, tablets, or computers.
- As the system grows, operational costs such as hosting and maintenance can be covered through service fees, subscription plans, or partnerships with restaurants.

3. Operational Feasibility

- Restaurants will benefit from better order management,
- Customers will enjoy faster and more convenient service.
- The system also supports for weak internet functionality, which helps users in areas with unstable internet connections.